FutureProof Financial

LMI & Portfolio Risk Reinsurance Program: Recommended Structure and Interaction

Based on the Equity Preservation Mortgage (EPM) v14c Model
GBM + Mean Reversion Equity Model (Shevchenko 2026)
50,000-path Monte Carlo Simulation | April 2025

Purpose: This paper sets out the recommended structure for the FutureProof Lenders Mortgage Insurance (LMI) program and its interaction with the Portfolio Risk Reinsurance program. It provides reinsurers with a comprehensive framework for structuring tail risk coverage for EPM portfolios.

> ***Important note on dollar amounts:*** *All dollar amounts in this paper are per-mortgage values for a representative $2,000,000 property at 80% LVR. These amounts will vary proportionally for different home values, LVR ratios, and annuity levels. They are not global constants.*

Contents

1. Executive Summary

The FutureProof Equity Preservation Mortgage (EPM) employs a two-layer insurance architecture designed to protect wholesale mortgage funders from credit losses. This paper addresses two key questions:

- **Part A:** How should the LMI program settings interact with the portfolio risk reinsurance program settings?
- **Part B:** How should a reinsurer structure the Portfolio Risk Reinsurance Program?

Key Findings (per-mortgage, $2M home at 80% LVR)

Metric	LMI Layer	Tail Risk Reinsurance	Combined
Insurance scope	100% of deficit paths	100% of deficit paths	100% of deficit paths
Covers deficits	Up to $547,901 per mortgage	Excess above $547,901 per mortgage	All deficits from $0
Paths with a claim	80% of deficit paths (smaller losses)	20% of deficit paths (larger losses)	100% of deficit paths
Probability of Claim	1.2%	0.24%	1.2%
Fair Premium (PV)	$1,716 per mortgage	$392 per mortgage	$2,108 per mortgage
Loaded Premium (50%)	$2,574 per mortgage	$587 per mortgage	$3,161 per mortgage
Premium timing	Upfront at origination	Upfront at origination	Upfront at origination

> ***Critical distinction:*** *The EPM is a mortgage, not a loan. Insurance claims can only arise at mortgage expiry (end of the 30-year term), not at any intermediate point. The insurance structure provides 100% coverage of all deficit outcomes from dollar zero — there is no deductible or uninsured gap. The two layers divide responsibility by loss severity: the LMI insurer covers smaller deficits, and the reinsurer covers the excess on larger deficits.*

Parameter Uncertainty and Three-Scenario Framework

The premium numbers above are produced under the model's base-case parameter set (MLE central estimates from historical data, 0.046% collar cost, base correlation). A companion paper — *Model Assumptions & Parameter Risk* — stress-tests each parameter against academic benchmarks (Shiller CAPE, Jordà-Schularick-Taylor, Black-Scholes collar pricing) and defines three scenarios that reinsurers should use side-by-side when structuring coverage:

Scenario	PoD (Yr 30)	Tail-layer claim rate	Use in reinsurance structuring
Base (model-as-written)	1.2%	0.24%	Quoted fair premium; pricing floor
Realistic-central	16.9%	~3.4%	Premium anchor; treaty rate-on-line

Scenario	PoD (Yr 30)	Tail-layer claim rate	Use in reinsurance structuring
Adverse-plausible	43.2%	~8.6%	Capital sizing; attachment/limit design; PML

Reinsurer guidance: *Structural features of the EPM (run-off mechanism, Payments Waterfall, ~21% loss severity given a deficit, 100% insurance from dollar zero, ~30-year float on upfront premium) are scenario-robust — they hold under every parameter set tested. What moves with the scenarios is expected loss frequency and, consequently, fair premium. Reinsurance pricing should anchor to the realistic-central scenario; capital and limit sizing should anchor to the adverse-plausible leg, not the base case.*

2. Stochastic Model Selection and Calibration

The EPM requires robust stochastic modelling of two key risk drivers over a 30-year horizon: equity market returns and interest rates. This section describes the model selection process, explains why each model was chosen, and details the parameter estimation methodology.

2.1 Interest Rate Model: Vasicek/Ornstein-Uhlenbeck Process

Cash rates are modelled using the Vasicek (Ornstein-Uhlenbeck) process, the standard model for mean-reverting interest rates. The exact discretisation (Shevchenko 2026, Section 1) is:

```
r(t+1) = r(t) * w + theta * (1 - w) + v * epsilon(t)
```

where w = exp(-kappa * dt), v = sigma * sqrt((1 - w^2) / (2 * kappa)), and epsilon ~ N(0,1).

This model captures the empirically observed tendency of interest rates to revert to a long-run mean, which is essential for realistic 30-year projections. A random walk model would produce unrealistic rate paths over such a long horizon.

2.2 Equity Model: GBM + Mean Reversion

Two candidate models were evaluated for equity returns using Maximum Likelihood Estimation (MLE) on historical S&P; 500 data (Shevchenko 2026, Sections 2-3):

Model	Equation	Log-Likelihood	Assessment
Geometric Brownian Motion (GBM)	S(t+1) = S(t)(1 + mu + sigma * epsilon)	-159.0	Standard model; no mean reversion; overstates 30-year tail risk
GBM + Mean Reversion (selected)	S(t+1) = S(t)(1 + mu + sigma * epsilon) + gamma * (M(t) - S(t))	-157.4	Superior fit; mean reversion reduces unrealistic extreme paths

The GBM + Mean Reversion model was selected because it provides a statistically superior fit (higher log-likelihood) and, critically, incorporates mean reversion towards a deterministic trend. The mean reversion component M(t) grows at the expected return rate, and the parameter gamma controls the speed at which the equity index reverts to this trend after deviations.

For a 30-year mortgage product, this is particularly important: pure GBM can generate extremely divergent paths over 30 years (both unrealistically high and unrealistically low), whereas the mean-reverting model produces paths that are more consistent with observed long-run equity market behaviour. This results in a more realistic — and somewhat lower — estimate of tail risk.

2.3 Parameter Estimation via Maximum Likelihood

All model parameters are estimated using Maximum Likelihood Estimation (MLE) on historical data — they are not arbitrary assumptions. The MLE procedure (detailed in Shevchenko 2026) provides:

- **Statistically optimal estimates:** MLE produces the parameter values that maximise the probability of observing the historical data under each model.
- **Standard errors:** The Fisher information matrix provides confidence intervals for each parameter, enabling sensitivity analysis.

- **Model comparison:** The log-likelihood values allow formal comparison between GBM and GBM+MeanRev to determine which model better fits the data.

Estimated Parameters (MLE)

Parameter	Symbol	MLE Estimate	Description
Equity expected return	mu_B	9.2% p.a.	Long-run S&P; 500 return
Equity volatility	sigma_B	16.6% p.a.	S&P; 500 annual volatility
Equity mean reversion speed	gamma_B	0.163	Speed of reversion to trend
Cash rate long-run mean	theta	2.13% p.a.	Long-run equilibrium cash rate
Cash rate mean reversion speed	kappa	0.24	Rate of mean reversion
Cash rate volatility	sigma_r	1.22% p.a.	Cash rate annual volatility
Equity-rate correlation	rho	0.30	Cross-correlation (appropriate for 30yr horizon)

The correlation of 0.30 between equity returns and interest rate innovations is estimated over the full historical sample. This moderate positive correlation is appropriate for the 30-year horizon of the EPM product.

2.4 Correlated Simulation Methodology

The Monte Carlo simulation generates 50,000 correlated paths for equity returns and interest rates over the 30-year mortgage term. Correlation is introduced via the Cholesky decomposition:

```
z1 ~ N(0,1) [equity innovation]
z2 = rho * z1 + sqrt(1 - rho^2) * z2_raw [rate innovation]
```

This ensures that equity and rate shocks are correlated within each simulated year, matching the observed empirical relationship. The full simulation tracks the investment account, mortgage balance, interest costs, fees, and the interest payment holiday mechanism year-by-year for each path.

2.5 Parameter Uncertainty and Alternative Calibrations

MLE point estimates on a ~75-year historical sample are the right starting point, but over a 30-year forward horizon each estimate carries material uncertainty. A separate *Model Assumptions & Parameter Risk* paper documents the stress-test fully; the short summary relevant to reinsurance structuring is:

Parameter	Base (MLE central)	Alternative / stress	Implication for tail-layer
Equity drift (mu_B)	9.2% p.a.	Shiller CAPE-implied 5–6% p.a.	Raises PoD materially; increases tail claim frequency
Mean reversion (gamma_B)	0.163	Bootstrap 95% CI [0.019, 0.447]	Lower gamma widens 30-year distribution; fatter tail

Parameter	Base (MLE central)	Alternative / stress	Implication for tail-layer
Cash rate long-run mean (theta)	2.13% p.a.	AU 30-yr MLE ~4.7% p.a.	Higher funding cost → higher mortgage balance at expiry
Equity-rate correlation (rho)	0.30	Crisis-period 0.5–0.7	Weakens hedge; drift/rate tail shocks become co-incident
Collar cost	0.046% p.a. net credit	Black-Scholes 0.16–0.44% p.a. net cost	Raises interest drag; increases deficit severity
Loss severity (structural)	~21% of deficit	Scenario-robust across stresses	Does not move materially — structural floor

Applying these stresses jointly produces the three scenarios flagged in the Executive Summary (base / realistic-central / adverse-plausible). Reinsurers should read the Executive Summary premium numbers as the base-case headline and use the realistic-central and adverse-plausible cases for pricing anchor and capital sizing respectively.

> ***What is scenario-robust:*** *the ~21% loss severity on deficit paths, the absence of prepayment risk (run-off mechanism), the cross-subsidy benefit at the portfolio level, the ~30-year float on upfront premium, and the zero-attachment-point structure of combined LMI + reinsurance. These are structural features of the product, not parameter-dependent.*

3. EPM Insurance Architecture Overview

The EPM insurance structure provides 100% coverage of all deficit outcomes from dollar zero. There is no deductible, no co-insurance gap, and no uninsured first-loss position for any party. The two insurance layers divide responsibility by **loss severity**:

- **LMI Layer:** Covers the full deficit for 80% of deficit paths (those with smaller deficits, up to the split boundary).
- **Tail Risk Reinsurance:** On the remaining 20% of deficit paths (those with larger deficits), the LMI insurer pays up to the split boundary and the reinsurer pays the excess above it.
- **Combined effect:** The wholesale mortgage funder bears zero credit risk — every deficit is fully covered from the first dollar.

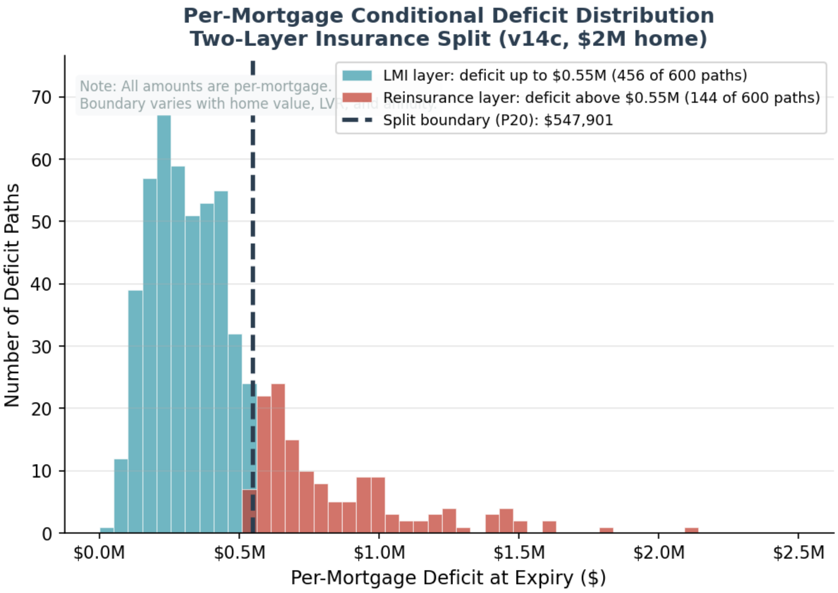
The Payments Waterfall

Before any insurance claim is triggered, a multi-step Payments Waterfall applies at the portfolio level. This waterfall dramatically reduces the actual claim frequency:

- **Step 1 — ETF Selldown:** The individual mortgage's investment account (S&P; 500 index-linked ETFs) is liquidated to cover the outstanding mortgage balance.
- **Step 2 — Cross-Subsidisation:** At the portfolio level, surpluses from performing mortgages are used to offset deficits from underperforming ones.
- **Step 3 — Net Deficit:** Only the residual deficit after Steps 1 and 2 triggers an insurance claim. This is the true PoC (Probability of Claim).

> *The Payments Waterfall reduces portfolio PoC from 1.2% (individual mortgage) to effectively zero (portfolio of 10,000+ EPMs) — the 109:1 surplus-to-deficit ratio ensures near-complete cross-subsidisation.*

Two-Layer Insurance Structure



The **split boundary** is set at the 20th percentile of the conditional deficit distribution for each mortgage. For the representative $2M home shown above, this boundary is $547,901 per mortgage. This boundary is not a global constant — it varies with home value, LVR, annuity amount, and mortgage term, and is recalculated via Monte Carlo simulation for each mortgage configuration.

- **LMI Layer:** Covers 80% of deficit paths in full (those with deficits below the split boundary). On the remaining 20% of paths, the LMI insurer pays up to the boundary amount.
- **Tail Risk Reinsurance:** On the 20% of deficit paths that exceed the split boundary, the reinsurer pays the excess above the boundary. The reinsurer has no liability on the other 80% of deficit paths.

> ***Key point for reinsurance underwriters:*** *Both insurers provide coverage from dollar zero. The split is by loss severity, not by attachment point. On every deficit path, the LMI insurer pays first (up to the boundary). On the 20% of paths where the deficit exceeds the boundary, the reinsurer pays the excess. There is no "attachment point" in the traditional XoL sense — the attachment is at zero for the combined structure.*

4. Part A: LMI Program — Structure and Settings

4.1 LMI Program Scope

The LMI program provides first-loss credit protection to the wholesale mortgage funder on every EPM originated. Unlike traditional LMI (which covers shortfalls from foreclosure sales), EPM LMI covers the deficit between the investment account value and the outstanding mortgage balance at mortgage expiry.

4.2 LMI Program Settings (per-mortgage, $2M home)

Setting	Value	Rationale
Coverage trigger	Mortgage expiry only (Year 30)	No intermediate claims; run-off mechanism prevents early exit
Split boundary	$547,901 per mortgage	P20 of conditional deficit distribution (recalculated per mortgage config)
Coverage scope	100% of deficit paths from $0	LMI pays full deficit on 80% of paths; pays up to boundary on remaining 20%
PoC (individual)	1.2%	Probability any individual mortgage results in a deficit
Conditional expected deficit	$333,305 per mortgage	Mean deficit given a deficit occurs
Fair premium (PV)	$1,716 per mortgage	Discounted expected loss at 2.13% (cash rate long-run mean)
Loaded premium (50%)	$2,574 per mortgage	Industry-standard loading for expenses + profit
Premium as % of max loan	0.16%	Competitive with traditional LMI at 80% LVR
Payment timing	Single upfront premium at origination	Funded from mortgage drawdown
Claim timing	At mortgage expiry (Year 30)	Deferred claims benefit insurer float

4.3 LMI Insurer Advantages

- **Deferred claims:** Claims only arise at mortgage expiry (Year 30). This provides ~30 years of investment float — the insurer collects the premium upfront and invests it for the full mortgage term before any claim liability crystallises.
- **Bounded exposure:** The split boundary caps the maximum LMI claim per mortgage at $547,901 (for a $2M home). Larger deficits are shared with the reinsurer.
- **No moral hazard:** The homeowner has no incentive to default — they retain full ownership of their property. The run-off mechanism prevents strategic early exit.
- **Transparent pricing:** Monte Carlo simulation with MLE-calibrated parameters provides precise, auditable premium calculations with quantified standard errors.

4.4 Interest Payment Holiday Mechanism

The EPM includes an interest payment holiday mechanism that protects the investment account during periods of underperformance. When the investment account falls below 90% of the initial mortgage amount, interest payments to the wholesale funder are suspended. Importantly:

- The **annuity payments to the borrower continue** during the holiday — it is the interest cost payments (to the funder) that pause.
- Deferred interest is accumulated and repaid when the investment account recovers above the exit threshold (145.8% of initial mortgage).
- This mechanism reduces the probability of deficit by preserving the investment account during downturns. While the average path has zero holidays, approximately 46% of individual paths enter holiday at some point, with a mean of 4.5 years across all paths.

5. Part B: Tail Risk Reinsurance — Structure and Settings

5.1 Reinsurance Program Scope

The Tail Risk Reinsurance program provides excess-of-boundary protection for the most severe deficit outcomes. It covers the 20% of deficit paths where the per-mortgage deficit exceeds the split boundary. On these paths, the reinsurer pays only the excess above the boundary — the LMI insurer still pays up to the boundary amount.

> ***Important clarification:*** *This is not a traditional excess-of-loss (XoL) treaty with an attachment point above zero. The combined LMI + reinsurance structure provides 100% coverage from dollar zero. The "split boundary" determines how much the LMI insurer pays on severe-deficit paths before the reinsurer becomes liable for the excess. The attachment point for the overall insurance structure is zero — no deficit goes uninsured.*

5.2 Reinsurance Program Settings (per-mortgage, $2M home)

Setting	Value	Rationale
Split boundary	$547,901 per mortgage	P20 of conditional deficit distribution — varies by mortgage configuration
Coverage	Excess above split boundary per mortgage	Reinsurer pays deficit minus boundary amount on severe paths
Probability of claim (individual)	0.24%	Only 0.24% of all mortgages trigger a reinsurance claim
Fair premium (PV)	$392 per mortgage	Discounted expected excess loss
Loaded premium (50%)	$587 per mortgage	Standard risk loading
Premium as % of max loan	0.0367%	Very small relative to mortgage size
Payment timing	Single upfront premium at origination	Funded from mortgage drawdown alongside LMI premium
Claim timing	At mortgage expiry (Year 30) only	Same deferred-claim benefit as LMI
Assessment basis	Per-mortgage	Each mortgage independently assessed at expiry

5.3 Why the Split Boundary Works

The 20th percentile of the conditional deficit distribution (P20) is the optimal boundary because:

- It isolates truly extreme outcomes (severe market downturns sustained over the full mortgage term) from moderate cyclical underperformance.
- 80% of deficit paths result in claims fully manageable by the LMI insurer (up to $547,901 per mortgage for a $2M home).
- The 20% tail represents only 0.24% of all mortgage paths — a low-frequency event well suited for reinsurance markets that specialise in severity risk.

- The reinsurance premium is economically efficient — only $392 fair premium per mortgage (0.04% of max loan) because claim probability is very low.
- The boundary is not a fixed dollar amount — it is recalculated for each mortgage configuration via Monte Carlo simulation, ensuring the split remains appropriate.

6. Part C: Interaction Between LMI and Tail Risk Reinsurance

6.1 Claim Flow Mechanics

The two programs interact through a clear, sequential claim process at mortgage expiry:

Step	Action	Responsible Party
1	Mortgage expires (Year 30) — investment account liquidated	FutureProof (servicer)
2	Payments Waterfall applied (ETF selldown, cross-subsidisation)	FutureProof (portfolio manager)
3	Net deficit determined after waterfall	FutureProof (servicer)
4a	If deficit <= split boundary: LMI pays full deficit (from $0)	LMI Insurer
4b	If deficit > split boundary: LMI pays up to boundary, reinsurer pays excess	LMI Insurer + Reinsurer
5	Mortgage funder made whole — zero credit loss	N/A

> ***Worked example (per-mortgage, $2M home):*** *If a mortgage expires with a $900,000 deficit (after the Payments Waterfall), the LMI insurer pays $547,901 and the reinsurer pays $352,099. If the deficit is $500,000, the LMI insurer pays the full $500,000 and the reinsurer pays nothing.*

6.2 Key Interaction Principles

Principle 1: No gap in coverage. The LMI layer and reinsurance layer together provide 100% coverage of all deficit outcomes from dollar zero. There is no deductible, no co-insurance, and no coverage gap. The mortgage funder bears zero credit risk.

Principle 2: Clear allocation by severity. The split is by loss severity, not by probability. Every deficit path is either: (a) fully covered by LMI (80% of deficit paths — smaller losses), or (b) split between LMI (pays up to boundary) and reinsurance (pays excess). The reinsurer only participates on the most severe 20% of deficit outcomes.

Principle 3: Independent pricing. Each layer is priced independently based on its own loss distribution. The LMI premium reflects expected losses up to the split boundary. The reinsurance premium reflects only the excess-of-boundary component.

Principle 4: Aligned incentives. The LMI insurer retains 80% of deficit frequency risk and pays on 100% of deficit paths (up to the boundary), ensuring active underwriting engagement. The reinsurer takes severity risk at low probability, which is their natural specialty.

6.3 Sensitivity of Boundary to Mortgage Configuration

The split boundary is calibrated to the P20 of the conditional deficit distribution for each mortgage configuration. The boundary is not a global constant — it shifts with the underlying risk profile of each mortgage:

Parameter Change	Effect on Split Boundary	Effect on Reinsurance Premium
Higher home value (e.g. $3M)	Increases proportionally (larger mortgage)	Increases proportionally
Higher equity volatility	Increases (wider deficit distribution)	Increases (more tail risk)
Lower equity return	Increases (more deficits, wider tails)	Increases moderately
Higher LVR	Increases proportionally	Increases roughly proportionally
Higher buffer cap	May increase or decrease (trade-off)	Marginal effect
Portfolio diversification	No direct effect (per-mortgage boundary)	Reduces aggregate exposure

7. Part D: Reinsurer Structuring Recommendations

7.1 Recommended Treaty Structure

The recommended structure for the Tail Risk Reinsurance program is a **per-mortgage severity-based reinsurance treaty** with the following key features:

Feature	Recommendation	Rationale
Treaty type	Per-mortgage excess-of-boundary	Reinsurer pays excess above split boundary on each mortgage
Split boundary	$547,901 per mortgage ($2M home)	P20 of conditional deficit — varies by mortgage config
Limit	Unlimited (or $3M per risk)	Tail events can produce large individual deficits
Term	30 years (matching mortgage term)	Claims deferred until mortgage expiry
Premium basis	Single upfront premium per mortgage	Paid at origination alongside LMI premium
Loaded premium	$587 per mortgage ($2M home)	Fair premium $392 + 50% loading
Aggregate limit (annual)	Optional — 10x expected annual claims	Catastrophe protection for extreme scenarios
Reinstatement	Automatic, unlimited	Continuous coverage essential for mortgage funder
Claims basis	Losses discovered at mortgage expiry	Triggered at Year 30

7.2 Portfolio-Level Considerations

The reinsurer should consider the following portfolio dynamics when structuring coverage:

- **Vintage diversification:** A portfolio of 1,000+ EPMs per year across multiple vintages means claims are spread over decades. A "bad" vintage (originated during a market peak) will have its claims offset by "good" vintages.
- **Geographic diversification:** FutureProof operates across US, Australia, NZ, and UK. Different equity markets and rate cycles provide natural diversification.
- **Deferred claims benefit:** The reinsurer collects the upfront premium and invests it for ~30 years before any claim liability emerges. The investment income on the accumulated premium reserve is substantial and should be factored into pricing.
- **Low frequency, moderate severity:** Only 0.24% of all mortgages trigger a reinsurance claim. When they do, the per-mortgage excess is bounded by the relationship between the split boundary and the maximum realistic deficit.

7.3 Alternative Treaty Structures

While per-mortgage severity-based reinsurance is the recommended primary structure, reinsurers may also consider:

Structure	Pros	Cons	Suitability
Per-mortgage excess-of-boundary (recommended)	Clean, transparent, simple pricing	Individual risk assessment needed	PRIMARY — best fit
Aggregate excess-of-loss	Protects against correlated losses	More complex pricing	COMPLEMENTARY — for catastrophe layer
Quota share	Proportional risk sharing	Less efficient for tail risk	NOT RECOMMENDED — wrong risk profile
Stop loss	Caps total loss ratio	Expensive for low-frequency risk	OPTIONAL — aggregate cap
ILW (Industry Loss Warranty)	Triggered by market index	Basis risk	INNOVATIVE — for systematic risk

7.4 Recommended Layered Structure for Larger Portfolios

For portfolios exceeding 5,000 EPMs, a layered reinsurance structure is recommended:

Layer	Type	Trigger	Limit	Estimated Rate
Layer 1: Working Layer	Per-mortgage excess	Per-mortgage deficit > $547,901	First $1M excess per mortgage	0.03% of outstanding
Layer 2: Buffer Layer	Per-mortgage excess	Per-mortgage excess > $1M above boundary	Next $2M excess per mortgage	0.015% of outstanding
Layer 3: Cat Layer	Aggregate excess	150% of expected annual claims	$10M xs attachment	0.005% of outstanding

This layered approach provides efficient risk transfer at each severity level and allows specialised reinsurers to participate in the layers that best match their risk appetite.

7.5 Capital Sizing Across Parameter Scenarios

The three-scenario framework from Section 2.5 maps directly onto reinsurance structuring decisions. Each scenario has a distinct role — the base case is a pricing floor, the realistic-central case is the pricing anchor, and the adverse-plausible case is the capital-sizing reference. Using only the base case would under-price the treaty and undersize capital; using only the adverse-plausible case would over-price and lose competitive traction.

Scenario	Tail-layer PoC	Expected loss (tail layer)	Reinsurer role
Base (model-as-written)	0.24%	~$392 per mortgage	Pricing floor — minimum acceptable premium
Realistic-central	~3.4%	~4–5x base expected loss	Rate-on-line anchor; reserve testing
Adverse-plausible	~8.6%	~10x base expected loss	Attachment/limit sizing; PML; capital allocation

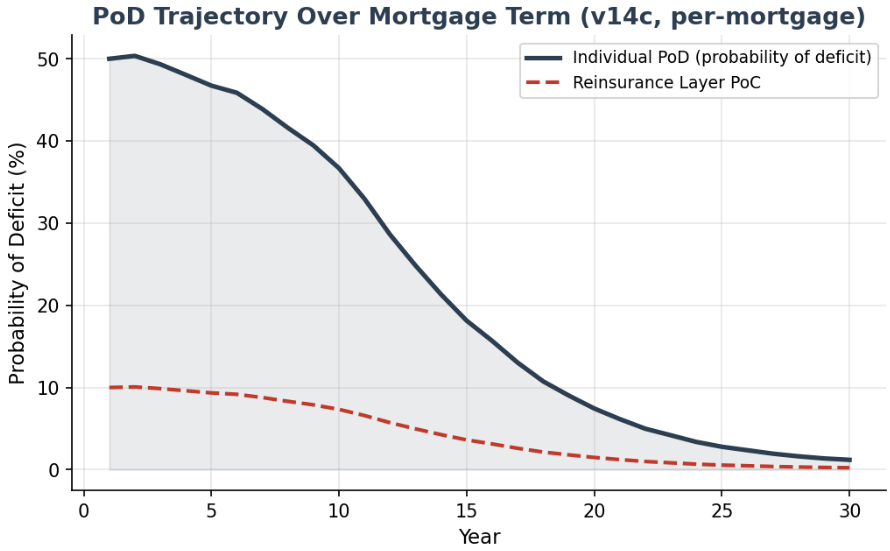
Practical implications for treaty design:

- **Premium loading beyond the stated 50%:** the 50% loading above is calibrated to the base-case fair premium. A reinsurer anchoring to realistic-central would need either a larger loading, a scenario-weighted fair premium, or explicit experience-rating triggers.
- **Per-risk limit:** should be sized against the adverse-plausible tail distribution, not the base-case tail. The maximum per-mortgage deficit observed under adverse-plausible parameters is the relevant severity reference.
- **Aggregate cat layer:** the optional aggregate limit (Section 7.1) becomes more valuable under adverse-plausible parameters, since claim frequency clusters more tightly around bad-vintage years. For adverse-plausible calibration, aggregate protection is strongly recommended rather than optional.
- **Experience-rating clauses:** given the wide spread between base and adverse-plausible, a profit-commission or swing-rated structure allows the reinsurer to write at realistic-central rates while sharing the benefit if actual experience tracks the base case.
- **Float economics are scenario-robust:** the ~30-year investment float on upfront premium is structural — it applies equally under every scenario. Even under adverse-plausible assumptions, the economic present value of the treaty remains attractive provided pricing reflects the scenario-appropriate expected loss.

8. Quantitative Analysis — v14c Model Results

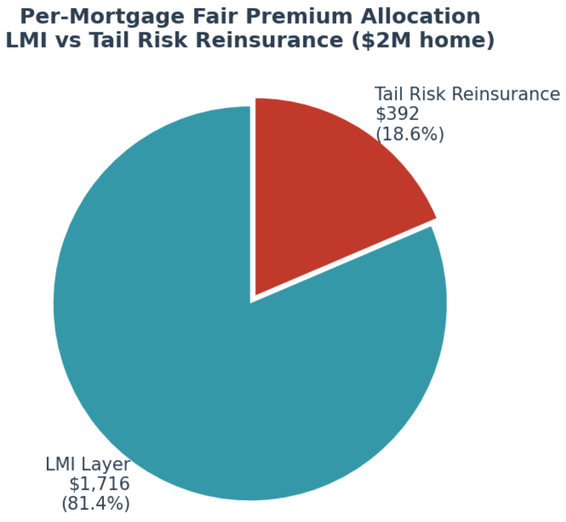
All values below are per-mortgage for a representative $2,000,000 property at 80% LVR.

8.1 PoD Trajectory Over Mortgage Term



The individual probability of deficit (PoD) declines from 50.0% in Year 1 to 1.2% at Year 30 as the PI amortisation progressively reduces the mortgage balance. The reinsurance layer PoC follows the same trajectory at approximately one-fifth of the individual PoD.

8.2 Premium Allocation (per-mortgage)



The total per-mortgage fair premium of $2,108 is split 81% / 19% between LMI and tail risk reinsurance. Both premiums are paid as single upfront amounts at mortgage origination, funded from the mortgage drawdown.

8.3 Surplus Distribution at Year 30 (per-mortgage)

Percentile	Surplus ($)	Interpretation
P1 (worst 1%)	$-55,115	Extreme downside — reinsurance claim likely
P5	$508,449	Significant deficit — within LMI/reinsurance coverage
P10	$808,302	Near boundary — moderate outcome
P25	$1,361,489	Moderate surplus — no claim
Median	$2,151,975	Typical outcome — healthy surplus
Mean	$2,435,992	Average outcome — strong surplus
P75	$3,194,847	Good outcome
P90	$4,419,493	Strong outcome
P99	$7,476,780	Exceptional outcome

9. Premium Pricing Framework

Both the LMI premium and the reinsurance premium are calculated using the same Monte Carlo simulation framework and are paid as single upfront amounts at mortgage origination.

9.1 LMI Premium Calculation (per-mortgage)

The LMI premium is calculated using the Discounted Expected Deficit methodology:

- **Step 1:** Run 50,000-path Monte Carlo simulation (GBM+MeanRev equity model, Vasicek rates, MLE-calibrated parameters) to generate surplus/deficit distribution at mortgage expiry.
- **Step 2:** Identify deficit paths (investment account < outstanding mortgage balance). Result: 1.2% of paths are in deficit.
- **Step 3:** Calculate the P20 of the conditional deficit distribution to establish the per-mortgage split boundary ($547,901 for a $2M home).
- **Step 4:** For each deficit path: LMI claim = min(deficit, split boundary). Discount each claim to present value at 2.13% (cash rate long-run mean).
- **Step 5:** Fair premium = mean of discounted claims = $1,716 per mortgage. Apply 50% loading for expenses and profit = $2,574 per mortgage.

9.2 Reinsurance Premium Calculation (per-mortgage)

The reinsurance premium follows the same methodology for the excess layer:

- For each deficit path exceeding the split boundary: reinsurance claim = deficit - split boundary.
- Discount each excess claim to present value at 2.13%.
- Fair premium = mean of discounted excess claims = $392 per mortgage. Apply 50% loading = $587 per mortgage.

9.3 Premium Payment Structure

Both the LMI premium and the reinsurance premium are structured as single upfront payments at mortgage origination, funded from the mortgage drawdown:

Premium Component	Amount (per-mortgage, $2M home)	Payment Timing	Funded From
LMI premium (loaded)	$2,574	Upfront at origination	Mortgage drawdown
Reinsurance premium (loaded)	$587	Upfront at origination	Mortgage drawdown
Total insurance cost	$3,161	Upfront at origination	Mortgage drawdown
As % of max loan	0.20%		

The upfront premium structure means the insurer and reinsurer both receive funds at origination and invest them for ~30 years before any claim liability arises. This investment float is a significant benefit and should be considered when evaluating the attractiveness of the risk.

10. Risk Transfer Summary

The complete risk transfer architecture ensures the wholesale mortgage funder bears zero credit risk. Insurance coverage begins at dollar zero — there is no uninsured first-loss position for any party:

Risk Layer	Coverage	Risk Bearer	Premium Basis
Portfolio cross-subsidy	Surplus mortgages offset deficit mortgages	FutureProof (portfolio manager)	None (portfolio structure)
First-loss insurance (mild deficit)	Deficit up to $547,901 per mortgage	LMI Insurer	$2,574 upfront per mortgage
Severity insurance (severe deficit)	Excess above $547,901 per mortgage	Reinsurer	$587 upfront per mortgage
Residual risk	None — full coverage from $0	N/A	N/A

> ***Key takeaway for reinsurers:*** *The EPM reinsurance opportunity is a severity-based risk with ~30 years of investment float on the upfront premium, zero prepayment risk, transparent MLE-calibrated pricing via Monte Carlo simulation, and natural portfolio diversification across vintages and geographies — all of which are scenario-robust. Expected claim frequency ranges from 0.24% (base case) to ~3.4% (realistic-central) to ~8.6% (adverse-plausible) per mortgage. Pricing should anchor to the realistic-central scenario; attachment, limit and capital should be sized to the adverse-plausible scenario.*

10.1 EPM vs Comparable Risk Profiles

To assist reinsurer underwriting assessment, we benchmark the EPM credit risk profile against the instruments most commonly seen in reinsurance portfolios.

Metric	EPM base	EPM realistic-central	Traditional LMI	Moody's AA Bond	AU RMBS (AA)
Cumulative tail-layer claim probability (30yr)	0.24%	~3.4%	2–4% of book	~1.5%	~1–2%
Loss severity (conditional on deficit)	~21%	~21%	15–25%	40–60%	Subordination-dependent
Expected loss (30yr, combined)	0.13%	~0.6–0.8%	0.6–1.25%	0.6–0.9%	0.05–0.15%
Claim timing	Year 30 only	Year 30 only	Peaks Year 3–7	Any year	Any year
Float on premium	~30 yrs	~30 yrs	~3–7 yrs	N/A	N/A
Correlation to trad. LMI	Near zero	Near zero	—	Low	Moderate

Metric	EPM base	EPM realistic-central	Traditional LMI	Moody's AA Bond	AU RMBS (AA)
Prepayment / lapse risk	None	None	Significant	Moderate	Significant

- **Scenario-sensitive vs scenario-robust metrics.** Expected loss and claim probability move materially across the three parameter scenarios (base → realistic-central roughly quadruples expected loss). Loss severity (~21%), claim timing (Year 30 only), float duration (~30 years), and the absence of prepayment risk are *structural* — they do not move with the parameter stresses. Reinsurer underwriting should separate these two categories.
- **Unique diversification value (scenario-robust).** Traditional LMI losses are driven by borrower unemployment, divorce, and rate stress — correlated with economic cycles. EPM losses are driven by 30-year equity index returns. This diversification benefit holds under every parameter scenario tested and is a genuine diversifier for reinsurance portfolios concentrated in traditional mortgage credit.
- **Exceptional float characteristics (scenario-robust).** Premiums are collected upfront at mortgage origination and claims can only arise at mortgage expiry. This ~30-year float applies under every parameter scenario — far longer than any traditional LMI product, where claims typically peak within 3–7 years. The economic value of the float is substantial even under adverse-plausible calibration.
- **Expected loss band, not single point.** The base-case combined expected loss of 0.13% would place the risk in AA–AAA territory. Under realistic-central parameters the expected loss is closer to 0.6–0.8%, roughly BBB-BB territory. Under adverse-plausible parameters it is higher still. Reinsurers should price to the realistic-central expected loss and hold capital against the adverse-plausible scenario.

10.2 Advantages for Each Participant

Participant	Key Advantage	Risk Retained
Mortgage Funder	Zero credit risk — full insurance coverage from $0	None (fully transferred)
LMI Insurer	First-loss insurer with bounded severity + 30yr float on upfront premium	Up to $547,901 per mortgage
Reinsurer	Low-frequency severity risk with 30yr float on upfront premium	Excess above $547,901 per mortgage
FutureProof	Origination fees + margin + profit share	Portfolio management (cross-subsidy)
Homeowner	Access to home equity while retaining ownership	None — investment account performance is insured

11. Appendix A: v14c Model Parameters (per-mortgage, $2M home)

Parameter	Value	Notes
Home value	$2,000,000	Representative Australian metropolitan property
Initial mortgage	$1,300,000	Effective LVR: 65.0%
Max LVR	80%	Max loan: $1,600,000
Mortgage type	Principal + Interest (30 years)	Builds to peak at Year 10, then amortises to zero
Annuity	$30,000/year	For first 10 years (paid to borrower)
Equity model	GBM + Mean Reversion	Shevchenko 2026, Section 3 — MLE calibrated
Equity return (mu_B)	9.2%	MLE estimate from S&P; 500 data
Equity volatility (sigma_B)	16.6%	MLE estimate
Mean reversion speed (gamma_B)	0.163	MLE estimate — pulls returns toward trend
Buffer cap	140%	Maximum annual return credited
Buffer floor	80%	Minimum annual return credited
Cash rate model	Vasicek/OU process	Shevchenko 2026, Section 1 — MLE calibrated
Cash rate (initial)	4.21%	Current RBA cash rate
Cash rate (long-run mean)	2.13%	MLE estimate (theta)
Cash rate speed (kappa)	0.24	MLE estimate — mean reversion speed
Cash rate vol (sigma_r)	1.22%	MLE estimate
Equity-rate correlation	0.3	MLE estimate — appropriate for 30yr horizon
Wholesale margin	2.0%	Funder borrowing cost above cash rate
Retail margin	0.70%	Lender margin
FP margin	0.50%	FutureProof origination margin
Hedging fee	0.25%	Volatility buffer cost
Collar price	0.046%	Net annual credit from cap/floor collar
LMI upfront	0.7%	Of max loan at origination
Tail risk reserve	0.05%	Annual tail risk provision
Profit share	10% every 5 years	Surplus drawn periodically
Interest holiday entry	0.75x initial mortgage	Interest payments pause below this
Interest holiday exit	1.458x initial mortgage	Interest payments resume above this
Simulation paths	50,000	SE on PoD: 0.05%

Source: FutureProofCalculator_Pavel_v14c (003).xlsm | monte_carlo_v14c_003_results.json

12. Appendix B: Stochastic Model Equations

The following equations are from Shevchenko (2026), "Parameter Estimation for Mean-Reverting Brownian Stochastic Model", which provides the mathematical framework and MLE methodology for all stochastic models used in the EPM simulation.

B.1 Cash Rate Model — Vasicek/OU Process (Section 1)

The cash rate follows an Ornstein-Uhlenbeck process with exact discretisation:

```
r(t+1) = r(t) * exp(-kappa * dt) + theta * (1 - exp(-kappa * dt))
+ sigma * sqrt((1 - exp(-2*kappa*dt)) / (2*kappa)) * epsilon(t)
```

where theta is the long-run mean rate, kappa is the mean-reversion speed, sigma is the volatility, and epsilon ~ N(0,1). The MLE estimates are: theta = 2.13%, kappa = 0.24, sigma = 1.22%.

B.2 Equity Model — GBM (Section 2)

The baseline Geometric Brownian Motion model for equity returns:

```
S(t+1) = S(t) * (1 + mu + sigma * epsilon(t))
```

MLE estimates: mu = 9.2%, sigma = 16.6%. Log-likelihood = -159.0.

B.3 Equity Model — GBM + Mean Reversion (Section 3, selected)

The selected model adds a mean-reverting component to GBM:

```
S(t+1) = S(t) * (1 + mu_B + sigma_B * epsilon(t)) + gamma_B * (M(t) - S(t))
```

where M(t) = M(0) * (1 + mu_B)^t is the deterministic trend (growing at the expected return rate), and gamma_B controls the speed of mean reversion. When S(t) falls below trend, the gamma_B term adds a positive correction; when S(t) is above trend, it adds a negative correction.

MLE estimates: mu_B = 9.2%, sigma_B = 16.6%, gamma_B = 0.163. Log-likelihood = -157.4 (superior to pure GBM).

B.4 Why Mean Reversion Matters for EPM

The mean reversion parameter gamma_B = 0.163 has a material impact on 30-year tail risk:

- Without mean reversion (pure GBM), equity paths can wander arbitrarily far from the expected trend over 30 years, producing unrealistically extreme outcomes in both tails.
- With mean reversion, paths that deviate significantly from the long-run trend experience a restoring force (gamma_B * (M(t) - S(t))) that pulls them back toward the trend.
- This is consistent with observed equity market behaviour: while short-term returns are unpredictable, long-run returns tend to revert toward historical norms.
- The practical effect is a lower probability of deficit (1.2%) than pure GBM models would produce — this is a more realistic, not more optimistic, estimate.

B.5 Correlation Structure

Equity and rate innovations are correlated using the Cholesky decomposition:

```
z_equity ~ N(0,1)
z_rate = rho * z_equity + sqrt(1 - rho^2) * z_independent
where rho = 0.30 (MLE estimate)
```

This ensures that equity and interest rate shocks within each simulated year reflect the observed positive correlation between the two risk drivers.

Reference: Shevchenko, P.V. (2026). "Parameter Estimation for Mean-Reverting Brownian Stochastic Model." Working paper prepared for FutureProof Financial.